

CIE

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TITLE: InventralQ: A Full-Stack AI-Driven Inventory Optimization System

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**DETAILS OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT
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ABSTRACT

1.1 Problem Background

The inventory management systems utilized by practically all small or medium-sized businesses today rely on either traditional paper-based processes or simple spreadsheet applications like Microsoft Excel. Both of these systems have large problems with inefficiency and common errors (i.e., incorrect records, delayed updates, and redundancies) throughout the business as a result of how they both operate. In almost every case, businesses will frequently encounter problems with stock shortages or overstocking, or they will lack visibility into their inventory movement, which will negatively impact their productivity and profitability.

Additionally, these types of systems do not provide the ability to support real-time updates and therefore can lead to challenges for multiple users trying to retrieve current data from various places. The solution to these issues is the creation of a Real time Inventory, which allows businesses to monitor, manage and update their Inventory in a timely and accurate manner.

Furthermore, there are many benefits to having real-time inventory management systems such as reduce error rates, create greater transparency about what inventory is on hand, provide accurate reports and decrease time spent searching for items, allow users to add items or make changes quickly and easily.

1.2 Objective of the Project

The primary purpose for InventraIQ is to create a unified web-based solution for businesses to manage and monitor their stock levels accurately at the same time. The system aims to minimize the potential for human error associated with manual data entry by automatically updating stock levels when various events occur (e.g., sale, purchase, return, or transfer) using automated processes instead.

In addition, the application provides an interface where multiple users can log in using their own unique user accounts with role-based access control to provide individual security throughout the application. The application is designed so that it can easily adapt to any organization's requirements for future growth through the ability for easy expansion while being user friendly at all stages of its development.

Finally, integrating predictive analytics through machine learning ('ML') technology into this platform will allow InventraIQ to use historical sales information (data) in conjunction with current market trends to help organizations better prepare for future demand by reducing their odds of understocking or overstocking their products. The project is focused on integrating machine learning methodologies to provide predictions of future demand for the next 30 days based on past sales data.

1.3 Proposed Solution

To simplify inventory replenishment, the proposed solution is a web-based, real-time inventory management system. The web-enabled real-time inventory management system will provide an internal interface for both Admin & User levels to use as they manage their products, to check stock levels and monitor & control inventory changes.

When added and sold within the Real-Time Inventory System, stock levels will reflect accurately (as changes are made) so that (automatic updates) are found in real-time - no manual tracking of your stock levels are necessary. This complete solution will consist of a full-stack solution comprised of an easy-to-use front-end, secure back-end and properly structured database.

In Addition, so that the Real-Time Inventory Management system becomes even more efficient and effective in fulfilling the current businesses' needs, we are implementing predictive analysis into the new system to enable more intelligent decision-making.

1.4 Key Features or Functionality

This system has multiple capabilities to enhance the management of inventory:

- Secure log-in using JWT authentication
- Role-based access control: Admin, Manager, Staff
- Manage products: add, modify, view, delete
- Real-time stock levels
- Low-stock alerts – reorder quickly
- Dead stock detection
- 30 days demand forecasting

1.5 Technology / Method Used

The system is developed using modern technologies:

Frontend	React.js	Used to build an interactive and responsive user interface
Backend Runtime	Node.js	Handles server-side execution and application logic

Backend Framework	Express.js	Used to create RESTful APIs and manage routing
Database	MongoDB	Stores user data, product details, and sales records
Authentication	JWT (JSON Web Token)	Provides secure user authentication and session management
Security	Bcrypt	Hashes passwords to ensure data security
Machine Learning Language	Python	Used to implement the prediction logic
ML Framework	Flask	Creates API endpoints for the prediction service
Version Control	Git & GitHub	Used for code management and collaboration
Development Tool	VS Code	Code editor used for development

1.6 Expected Outcome & Benefits

The system should provide:

- Centralised inventory control
 - Real time inventory updates
 - Automatic transaction recording
 - Better accuracy and less manual errors
 - More informed decisions made possible by using data-driven insights
- Also, a machine-learning capability will assist in planning inventory for better management and stock issues.

The machine learning feature further helps in planning inventory effectively and avoiding stock-related issues.

1.7 Original Contribution Statement

Using real-time tracking and predictive analytics, InventraIQ is improving the management of inventory. Some critical contributions include:

- Transparency and accuracy have both improved
- Zero-data handling in AI prediction model to provide increased accuracy in sales when there are no sales at any given time, on specific days.
- Stock planning is now based on data
- Multi-user role-based access control for commonly utilized operations to increase safety, security & reliability.

By introducing these technological features, inventory systems are going to become more efficient and practical than previous methods.

Machine Learning Integration

2.1 Use of Machine Learning in the System

Machine learning is applied to help improve the management of inventory at InventraIQ. By reviewing historical sales data, they can determine trends in customer buying patterns and demand. Machine learning techniques, such as linear regression, allow them to predict future demand based upon the previous demand trends for a specific item.

By providing this predictive data, they can assist businesses in making better and more informed decisions regarding inventory, such as:

- Recommendations for when to restock items
- Fast-moving vs. slow-moving items
- Low-demand item
- Identify Dead Stock
- The system includes the capability to predict the anticipated demand for the next 30 days making it easier for businesses to plan their inventory in advance

Using this information helps to maximize the efficiency of inventory management and decision making for businesses.

System Architecture and Workflow

3.1 System Architecture

Frontend: React.js (User Interface)

Backend: Node.js with Express.js

- Auth Middleware
- Role Middleware
- Product Controller
- API Routes

Database: MongoDB (Users, Products, Sales Data)

Machine Learning Service: Flask (Demand Prediction using Python)

3.2 Work-flow of InventralQ

InventralQ starts with user identification – logging in or registering securely in the system. Based on their assigned role (Admin/Manager/Staff), users receive appropriate level access to perform tasks/functionalities in the system. Following a successful login, the user is presented with a centralized dashboard that provides visibility to all products in stock.

User products may be managed per the functions available to the user role as outlined above (e.g. Add New Items to Inventory, Update Product Details). After a sale has been recorded against a product in the system, the inventory will reduce the appropriate stocked item quantity accordingly and record the sale transaction details in the database.

All information related to Users, Products, Sales are stored in a MongoDB database. Recorded Sale data is sent to a Flask-based ML service on a periodic basis for analysis & prediction using data analytical methods and a linear regression model that analyzes historic sales patterns to predict sales demand over the next 30 days. Based upon these predicted quantities, users will receive valuable information regarding inventory replenishment recommendations, identifying fast/moving items, and identifying dead/low demand products, thus, empowering users to make smart inventory management & planning decisions.

3.3 Module Description

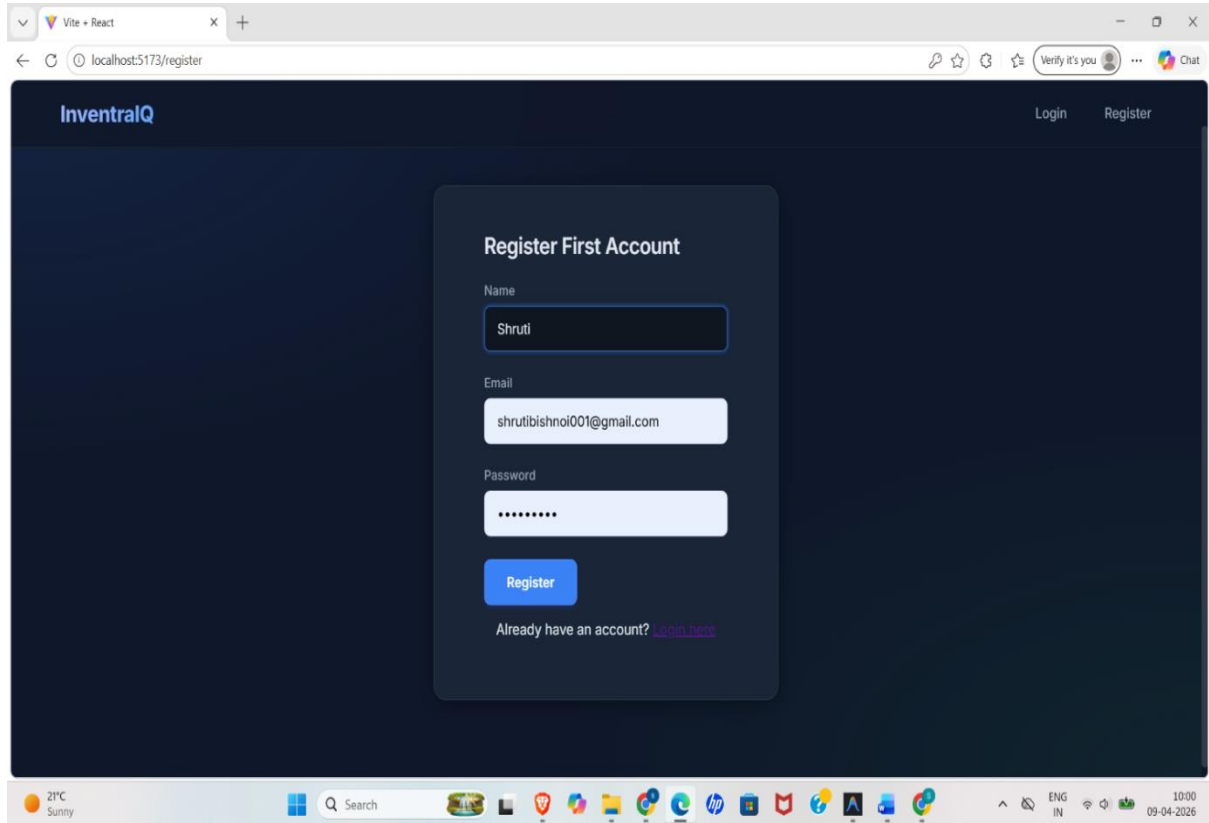
The system is designed based on the MVC (Model-View-Controller) architecture. It can also be viewed in terms of functional modules. which represent logical separations of the systems features.

The modules serve the following purposes:

- User Management module: Registering users, logging into the system, managing user profiles, providing role-based access (Admin, Manager, Staff).
- Product Management module: Add/update/delete products; add categories to products; add/update product details, including supplier.
- Inventory Processing module: Track stock levels in real-time; update stock levels (Adding/Selling/Returning); adjust inventory.
- Sales & Transaction module: Record sales/purchase transactions; automatically update inventory upon sale or purchase; keep track of historical sales and purchases.
- Alert module: Show alerts when stock item levels are low; set thresholds for showing alerts.
- ML Demand Prediction Module: Historical sales data will be used to forecast demand by predicting what future stock requirements will be and recommending reorder points, as well as performing trend analysis on sales.
- Admin module: Provide a central location to manage users, products, inventory, and system.

Preview

User Register



The screenshot shows a web browser window with the URL `localhost:5173/register`. The page features the **InventralQ** logo in the top left and navigation links for **Login** and **Register** in the top right. The main content is a dark-themed registration form titled **Register First Account**. It contains three input fields: **Name** (with the value "Shruti"), **Email** (with the value "shrutibishnoi001@gmail.com"), and **Password** (masked with dots). Below these fields is a blue **Register** button and a link that says "Already have an account? [login here](#)". The browser's taskbar at the bottom shows a weather widget (21°C Sunny), a search bar, and various application icons.

InventralQ Login Register

Register First Account

Name

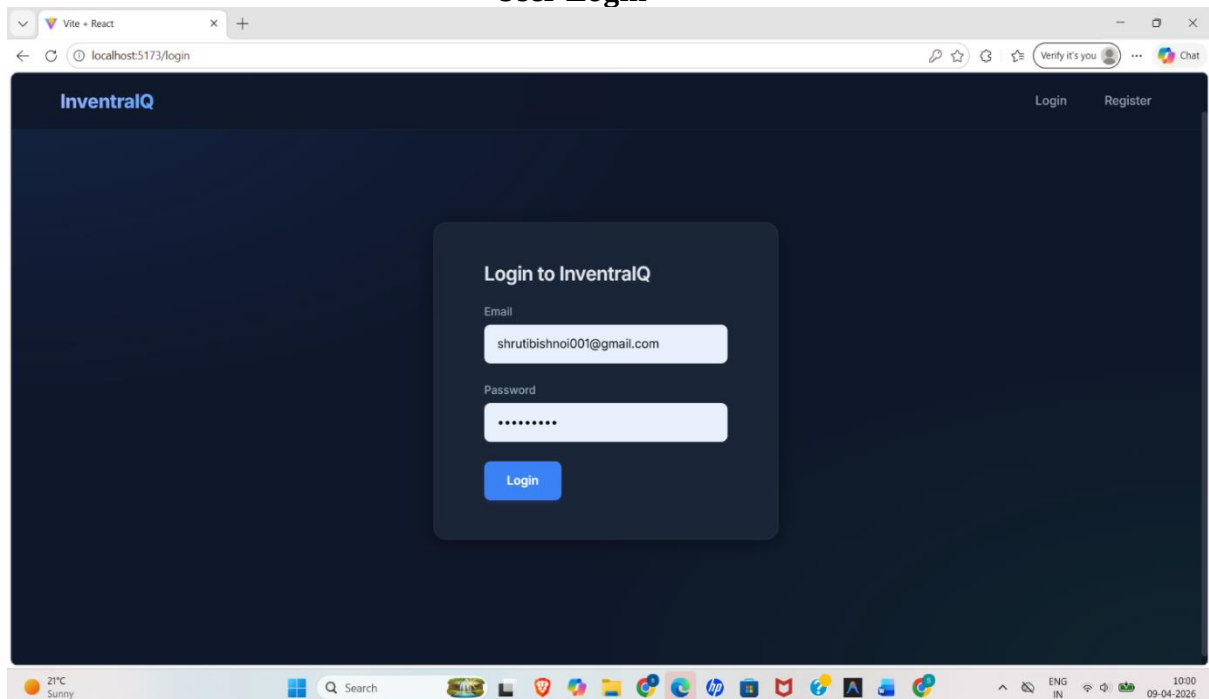
Email

Password

Register

Already have an account? [login here](#)

User Login



The screenshot shows a web browser window with the URL `localhost:5173/login`. The page features the **InventralQ** logo in the top left and navigation links for **Login** and **Register** in the top right. The main content is a dark-themed login form titled **Login to InventralQ**. It contains two input fields: **Email** (with the value "shrutibishnoi001@gmail.com") and **Password** (masked with dots). Below these fields is a blue **Login** button. The browser's taskbar at the bottom shows a weather widget (21°C Sunny), a search bar, and various application icons.

InventralQ Login Register

Login to InventralQ

Email

Password

Login

Role based access

The screenshot shows a web browser window with the URL `localhost:5173/register`. The page features a dark blue header with the **InventralQ** logo on the left and navigation links on the right: `Hello, Shruti (admin)`, `Admin Dashboard`, `AI Insights`, `Register User`, `Make a Sale`, and `Logout`. A user profile icon with the text `Verify it's you` is also present. The main content area contains a **Register New User** form with the following fields:
- **Name**: A text input field containing `Shruti`.
- **Email**: A text input field containing `shrutibishnoi001@gmail.com`.
- **Password**: A password input field with masked characters `.....`.
- **Role**: A dropdown menu with `Admin` selected. The dropdown list shows `Staff`, `Manager`, and `Admin` as options.
The Windows taskbar at the bottom shows the date `09-04-2026` and time `10:04`.

Admin Dashboard

The screenshot shows a web browser window with the URL `localhost:5173/admin`. The page features a dark blue header with the **InventralQ** logo on the left and navigation links on the right: `Hello, Shruti (admin)`, `Admin Dashboard`, `AI Insights`, `Register User`, `Make a Sale`, and `Logout`. A user profile icon with the text `Verify it's you` is also present. The main content area has the title **Admin Dashboard** and three sub-sections: `Inventory Management` (highlighted with a blue underline), `Staff Controls`, and `Sales Ledger`. Below these is a **Add New Product** form with the following fields:
- **Product Name**: A text input field.
- **Category**: A text input field.
- **Price (\$)**: A text input field.
- **Cost Price (\$)**: A text input field.
- **Current Stock**: A text input field.
- **Min Stock Level**: A text input field.
- **Supplier Name**: A text input field.
The Windows taskbar at the bottom shows the date `09-04-2026` and time `10:00`.

Manager Dashboard

The screenshot shows the Manager Dashboard for the InventaIQ application. The browser address bar indicates the URL is localhost:5174/admin. The dashboard header includes the InventaIQ logo, a user greeting "Hello, Lavanya (manager)", and navigation links for "AI Insights", "Make a Sale", and "Logout". The main content area features a "Manager Dashboard" title and three tabs: "Inventory Management" (selected), "Restock Inventory", and "Sales Ledger". Below the tabs is a form titled "Add New Product" with input fields for "Product Name", "Category", "Price (\$)", "Cost Price (\$)", "Current Stock", "Min Stock Level", and "Supplier Name". The Windows taskbar at the bottom shows the date as 26-04-2026 and the time as 11:58.

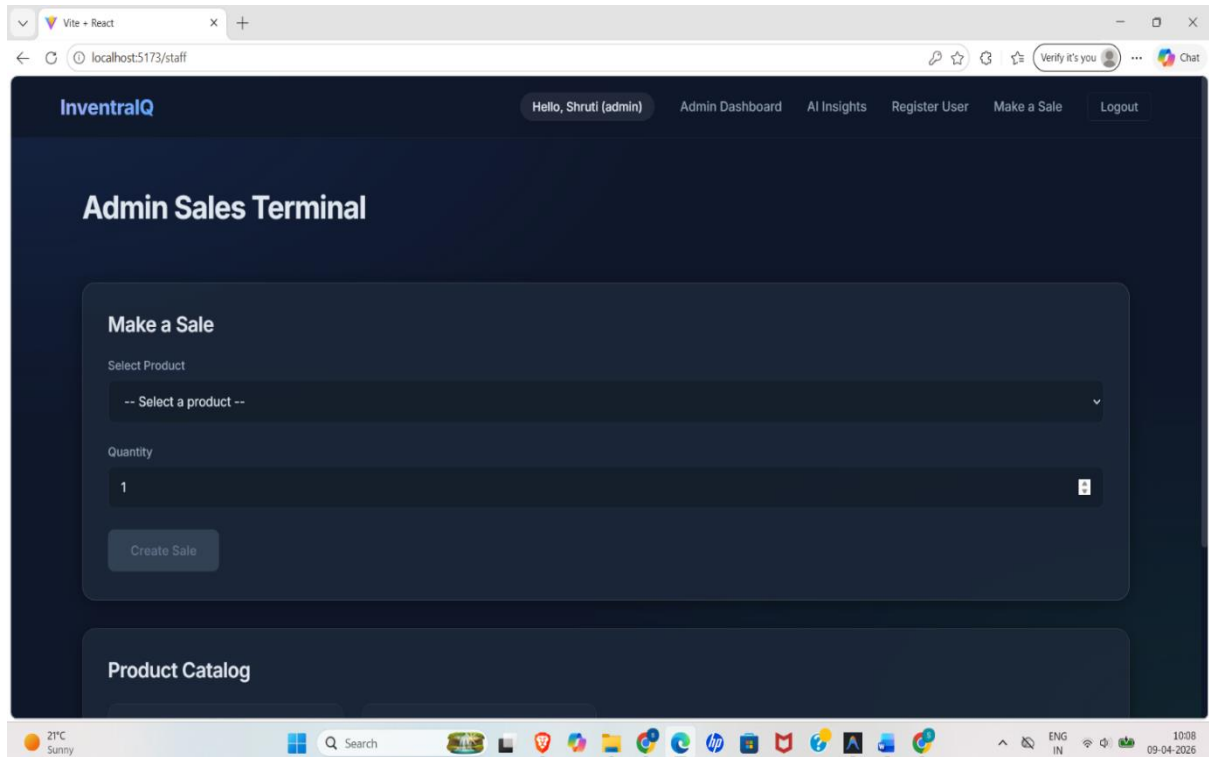
Staff Control

The screenshot shows the Admin Dashboard for the InventaIQ application. The browser address bar indicates the URL is localhost:5173/admin. The dashboard header includes the InventaIQ logo, a user greeting "Hello, Shruti (admin)", and navigation links for "Admin Dashboard", "AI Insights", "Register User", "Make a Sale", and "Logout". The main content area features an "Admin Dashboard" title and three tabs: "Inventory Management", "Staff Controls" (selected), and "Sales Ledger". Below the tabs is a section titled "Staff Overview" containing a table with staff information.

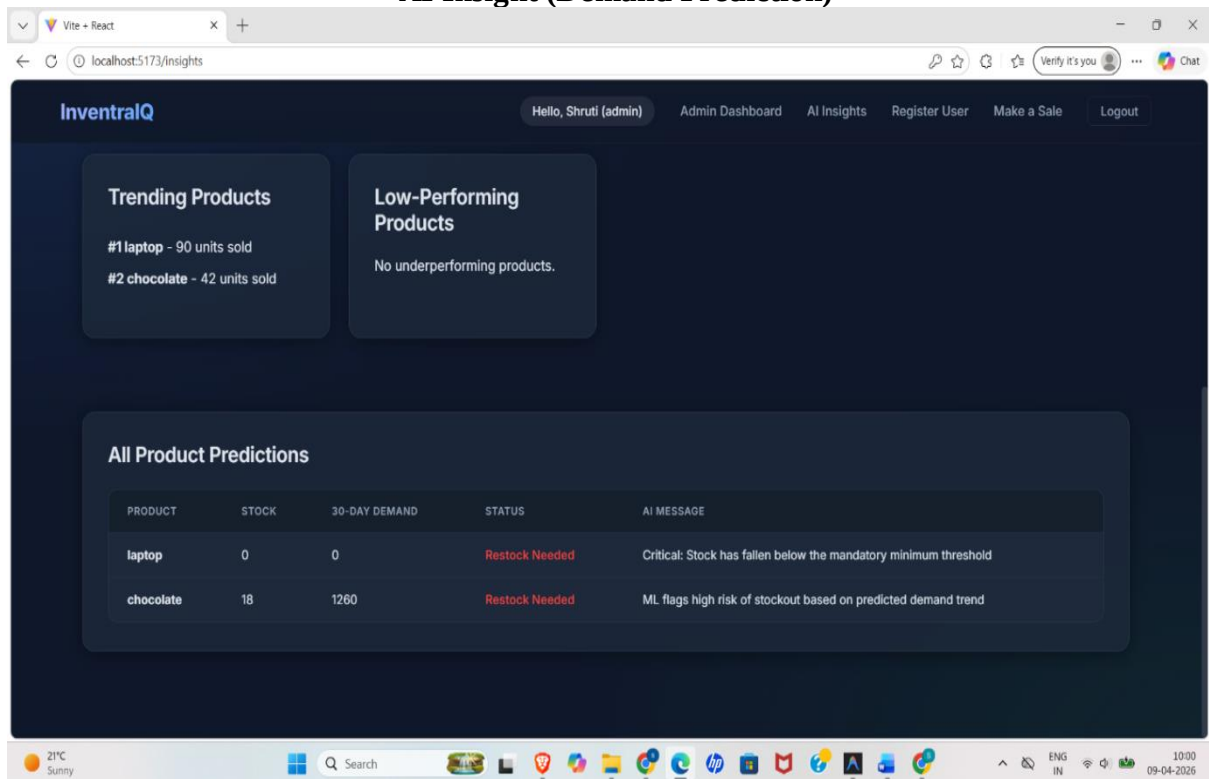
NAME	EMAIL	ROLE	ACTIONS
kannan	kannan@gmail.com	Manager	Delete

The Windows taskbar at the bottom shows the date as 09-04-2026 and the time as 10:00.

Sales Terminal



AI Insight (Demand Prediction)



Dead Stock

The screenshot displays the InvenIQ Admin Dashboard in a web browser. The browser's address bar shows the URL `localhost:5174/insights`. The dashboard's header includes the InvenIQ logo and a navigation menu with links for 'Hello, Shruti (admin)', 'Admin Dashboard', 'AI Insights', 'Register User', 'Make a Sale', and 'Logout'. The main content area is divided into several sections: a 'Priority Restock Recommendations' box stating 'No immediate restocks required based on current predictions.', a 'Dead Stock Warning' box with a yellow warning icon and the text 'Awesome! The AI detects no completely stagnant stock in your warehouse.', a 'Trending Products' box listing '#1 laptop - 120 units sold' and '#2 chocolate - 103 units sold', and a 'Low-Performing Products' box stating 'No underperforming products.' The Windows taskbar at the bottom shows the system clock as 16:05 on 25-04-2026, along with weather information (39°C Sunny) and various application icons.

InvenIQ

Hello, Shruti (admin) | Admin Dashboard | AI Insights | Register User | Make a Sale | Logout

Priority Restock Recommendations

No immediate restocks required based on current predictions.

Dead Stock Warning

Awesome! The AI detects no completely stagnant stock in your warehouse.

Trending Products

- #1 laptop - 120 units sold
- #2 chocolate - 103 units sold

Low-Performing Products

No underperforming products.

39°C Sunny | Search | 16:05 25-04-2026

References

- React – A JavaScript library for building user interfaces.
- Available at: <https://reactjs.org/>
- Node.js – JavaScript runtime environment.
- Available at: <https://nodejs.org/>
- Express.js – Web application framework for Node.js
- Available at: <https://expressjs.com/>
- MongoDB – NoSQL database for modern applications.
- Available at: <https://www.mongodb.com/>
- JSON Web Tokens (JWT) – Secure authentication method.
- Available at: <https://jwt.io/>
- Flask– A lightweight WSGI web framework. Pallets Projects.
- Available at: <https://flask.palletsprojects.com>